

GMMamba: Group Masking Mamba for Whole Slide Image Classification

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Abstract

Recent advances in selective state space model (Mamba) have shown great promise in whole slide image (WSI) classification. Despite this, WSIs contain explicit local redundancy (similar patches) and irrelevant regions (uninformative instances), posing significant challenges for Mamba-based multi-instance learning (MIL) methods in capturing global representations. Furthermore, bag-level approaches struggle to extract critical features from all instances, while group-level methods fail to adequately account for tumor dispersion and intrinsic correlations across groups, leading to suboptimal global representations. To address these issues, we propose group masking Mamba (GMMamba), a novel framework that combines two elaborate modules: (1) intra-group masking Mamba (IMM) for selective instance exploration within groups, and (2) cross-group super-feature sampling (CSS) to ameliorate long-range relation learning. Specifically, IMM adaptively predicts sparse masks to filter out features with low attention scores (i.e., uninformative patterns) during bidirectional Mamba modeling, facilitating the removal of instance redundancies for compact local representation. For improved bag prediction, the CSS module further aggregates sparse group representations into discriminative features, effectively grasping comprehensive dependencies among dispersed and sparse tumor regions inherent in large-scale WSIs. Extensive experiments on four datasets demonstrate that GMMamba outperforms the state-of-the-art ACMIL by 2.2% and 6.4% in accuracy on the TCGA-BRCA and TCGA-ESCA datasets, respectively.

1. Introduction

Histopathology slides are indispensable for cancer diagnosis, providing critical insights into tissue morphology and cellular structures [3, 9, 30]. However, the complexity of biological structures in these slides complicates the detection of subtle pathological variations [26, 34]. Recent

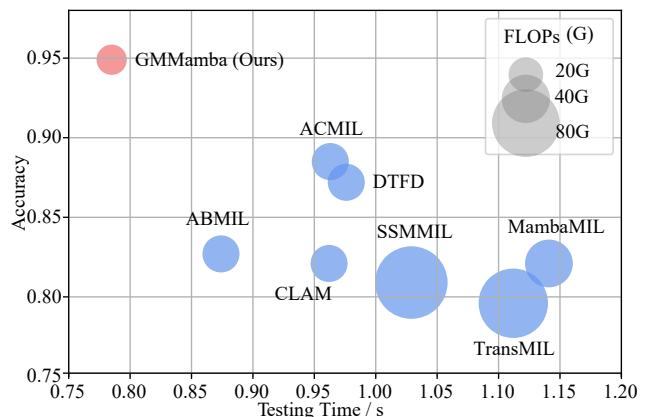


Figure 1. Comparative results of FLOPs, testing time, and accuracy with representative MIL methods on the TCGA-ESCA dataset. Our GMMamba achieves superior performance with higher efficiency. The size of each circle represents the FLOPs.

advances in digital pathology have enabled the conversion of physical slides into high-resolution whole slide images (WSIs), creating new opportunities for deep learning-driven tumor diagnosis. Despite this progress, the gigapixel resolution and diverse tumor patterns inherent in WSIs make pixel-level manual annotation prohibitively laborious and costly [26, 29, 38]. To alleviate these limitations, multi-instance learning (MIL) addresses this challenge by segmenting WSIs into patches (instances) and leveraging weak bag-level labels for analysis [7, 16, 32, 48, 50]. In general, previous efforts are rooted in exploring instance correlations to construct a global representation for bag prediction.

A mass of studies improve bag-level representation, including Mean-Pooling, attention mechanisms [1, 17, 39, 51], recurrent and graph neural networks [3, 28], and Transformer [29, 31]. Despite their considerable success, these methods face a critical trade-off: computational efficiency versus discriminative bag representation, especially in WSIs with numerous instances and complex tissue patterns. As shown in Figure 1, Transformer-based MIL methods (e.g., TransMIL [31]) suffer from high computational

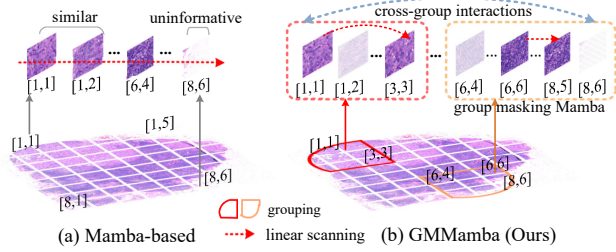


Figure 2. Instances in WSI exhibit explicit local redundancies (*i.e.*, similar and uninformative instances). Our GMMamba enhances the efficiency of Mamba-based MIL methods through intra-group masking Mamba (IMM) and cross-group super-feature sampling (CSS) modules, facilitating the exploration of crucial instances and global representations.

costs when handling massive instances, resulting in suboptimal performance and prolonged inference time.

Recently, the emergence of the selective state space model (Mamba) has driven significant progress in modeling long sequences with linear complexity [23, 24, 40, 42, 52]. Despite this advantage, two key limitations impede its application to WSI analysis: local redundancy and suboptimal global representation. **1) Redundant Local Modeling:** Bag-based Mamba processes all instances uniformly, leading to unnecessary computational overhead from non-discriminative features and potential loss of critical diagnostic information, as illustrated in Figure 2. **2) Sparse Global Representation:** Tumor regions in WSIs exhibit spatial dispersion and sparsity, complicating the aggregation of discriminative features across groups. Current methods inadequately model inter-group correlations, impairing global representation.

To alleviate these issues, prior attempts have extended the MIL framework by applying random masking [14, 30] or instance selection strategies [13, 20, 47]. However, these approaches often sacrifice critical information or introduce training instability. While feature similarity and attention-based selection improve bag representation, their computational costs remain prohibitive for large-scale WSI analysis. Moreover, existing approaches overlook the spatial dispersion of tumor instances across groups.

This raises a pivotal question: *How to harmonize the advantages of Mamba with local similarity in WSI to eliminate redundancy and ameliorate global representations for more accurate and efficient bag-level label inference?*

To answer this question, this paper proposes a novel group masking Mamba (GMMamba) for WSI classification. GMMamba mitigates undesirable local redundancy through an intra-group masking Mamba (IMM) and enhances global representation via cross-group super-feature sampling (CSS) modules. Unlike previous works that perform random groupings, GMMamba employs a location-based clustering scheme to group semantically related in-

stances, reducing training uncertainty. To eliminate redundant and irrelevant instances within the group, IMM adaptively incorporates learnable sparse masks between bidirectional Mamba (BiMamba) blocks to dynamically prune low-attention features. This enables Mamba to focus on salient instances while enjoying the merit of linear scanning, resulting in efficient yet compact local representations. For improved correlation learning across groups, CSS aggregates discriminative features from spatially dispersed tumor regions by sampling “super-features” from group representations. Moreover, to further enhance bag prediction, we generate the class token using multi-head attention mechanisms, further boosting the robustness of the global bag representation.

Our main contributions are summarized as follows:

- A novel GMMamba framework that harmonizes Mamba’s efficiency with group masking and cross-group interaction to address redundancy and sparse tumor dispersion in WSIs.
- An intra-group masking Mamba (IMM) is devised to incorporate location-based grouping and feature selection into Mamba for efficient and compact exploration of instance relationships. In addition, we pioneer a cross-group super-feature sampling (CSS) module to aggregate sparse and dispersed tumor features across groups, leading to a more accurate and comprehensive bag representation.
- Extensive experiments on public datasets demonstrate the superiority of our GMMamba over state-of-the-art methods, achieving accuracy improvements of 2.2%, 6.4%, 3.9%, and 1.0% on the TCGA-BRCA, TCGA-ESCA, BRACS, and TCGA-Lung datasets, respectively, while enjoying faster inference times.

2. Related Work

This section discusses relevant advancements related to our GMMamba, including multi-instance learning, state space models, and data redundancy optimization.

2.1. Multi-instance Learning

MIL [18] aims to predict WSI labels by exploring relationships among instances and forming global representations for bag-level label prediction. Early approaches often used Mean-Pooling and attention-based weighting to aggregate instances [1, 17]. However, due to the sparse nature of tumor instances in large-scale WSIs, these methods struggle to generate discriminative global representations [7, 35, 50]. While advanced technologies, *e.g.*, graph neural networks [5, 13] and Transformers [31, 37, 39, 44], have improved the capability of global context modeling,

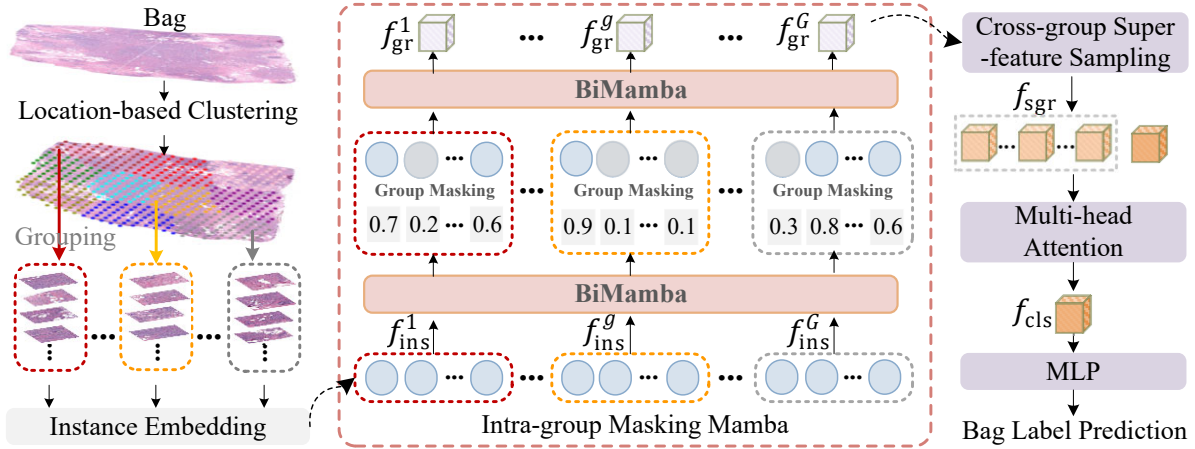


Figure 3. The architecture of our proposed group masking Mamba network (GMMamba) for WSI classification. GMMamba employs a location-based clustering scheme to divide a bag into multiple groups $\{f_{\text{ins}}^g\}_{g=1}^G$. The intra-group masking Mamba (IMM) module then predicts attention masks for a bidirectional Mamba (BiMamba) to eliminate redundant instances, facilitating more compact local representations $\{f_{\text{gr}}^g\}_{g=1}^G$. Subsequently, a cross-group super-feature sampling (CSS) module captures discriminative features from $\{f_{\text{gr}}^g\}_{g=1}^G$ to generate comprehensive super-feature group representations f_{sgr} . Finally, the class token f_{cls} is aggregated with f_{sgr} using multi-head attention mechanisms for final bag-level label prediction.

they are hindered by limited performance and high computational costs [41, 45]. Recently, Mamba has emerged as an effective yet efficient alternative for capturing long-range dependencies with linear computational complexity [23, 42]. Despite these advancements, previous MIL-based methods often overlook the undesirable local redundancy within WSIs [6, 20]. Some approaches attempt to mitigate this issue by selecting key features, with typical efforts in feature clustering, attention scores, and pooling [13, 34, 53]. Nevertheless, they require elaborated modules to identify non-informative patterns, making them less efficient for large-scale WSIs. To overcome these challenges, we propose GMMamba, which effectively reduces redundancy while efficiently modeling long-range dependencies among instances, leading to improved performance and computational efficiency.

2.2. State Space Model

The state space model (SSM) [33] excels at modeling long-range dependencies but suffers from input-invariant parameterization, limiting its adaptability to dynamic or spatial patterns. To address this limitation, Mamba [11] introduces selective scanning mechanisms and hardware-aware algorithms, achieving improved performance with linear complexity. These advancements have spurred applications in visual tasks [23, 27, 40, 41], where techniques such as bidirectional, four-way, multi-scale, and local-based scanning [15, 24, 42, 54] enhance global representation learning. However, non-unidirectional methods introduce redundant computations and feature interference [19, 49], particularly in WSIs with a massive number of redundant instances. To address this, our GMMamba employs a sparse instance masking scheme, selectively modeling

crucial long-range instances while eliminating redundancy, thereby achieving efficient instance-aware exploration.

2.3. Data Redundancy Optimization

Reducing data redundancy has drawn widespread attention across various vision and text tasks by eliminating similar or irrelevant data, which in turn improves model performance [14, 20, 34]. To reduce redundancy, previous approaches primarily focused on two strategies: data dropping and similarity replacement [6, 30, 47]. The former randomly discards a proportion of the data, while the latter employs distance metrics and clustering to replace similar data with representative centers [30, 47]. However, both methods risk discarding crucial details, potentially undermining accuracy and stability. To address this, attention- and pooling-based feature selection methods have emerged as popular solutions [1, 13] for filtering out irrelevant features. While they are effective at reducing redundancy, their performance often deteriorates when applied to large-scale and complex datasets, WSIs in particular, where identifying truly informative features becomes increasingly challenging. To this end, our proposed IMM effectively evicts similar and uninformative instances by incorporating attention-based sparse masks with Mamba, facilitating the representation of key instances for accurate label predictions.

3. Method

This section begins with the formulation of MIL and SSM, followed by an overview of the proposed GMMamba network. It then provides a detailed explanation of the intra-group masking Mamba (IMM) and cross-group super-feature sampling (CSS) modules.

3.1. Preliminaries

MIL Formulation. Given a WSI $\{X, Y\}$, we divide the non-background tissue regions X into B non-overlapping instances $\{(x_b \in R^{W \times H \times 3}, y_b) \mid 1 \leq b \leq B\}$ as a “bag”, where Y , H , W and y_b denote the bag label, height, width and unknown class label of the instance x_b , respectively. To enable cost-effective training, the classical pipeline for MIL-based WSI analysis embeds an instance x_b into a 1D-dimensional feature vector f_{ins}^b using pre-trained encoders [26, 44, 50]. Most existing studies [1, 31, 39] explore the correlations among $\{f_{\text{ins}}^b\}_{b=1}^B$ using an aggregator ϕ_{aggr} to form a global bag representation, followed by a multi-layer perceptron (MLP) to predict the bag label $\hat{Y}$. In the case of a binary classification task, as expressed in Eq. (1), if $\hat{Y} = 1$, the bag is labeled as “positive”; otherwise, it is labeled as 0 (negative).

$$Y = \begin{cases} 1, & \text{if } \hat{Y} = \text{MLP}(\phi_{\text{aggr}}(\{f_{\text{ins}}^b\}_{b=1}^B)) : \hat{Y} = 1 \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

State Space Model. The SSM [33] maps input sequences $x(t)$ at time t to output sequences $y(t)$ through latent states $h(t)$ using a deep learning model. This transformation is mathematically formulated as linear ordinary differential equations:

$$h(t) = \bar{\mathbf{A}}h(t-1) + \bar{\mathbf{B}}x(t), \quad (2)$$

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad (3)$$

$$y(t) = \mathbf{C}h(t) + \mathbf{D}x(t), \quad (4)$$

where $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are learned parameters, and $\mathbf{D}$ serves as a skip connection. $h'(t)$ denotes the derivative of $h(t)$. $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ are the discrete counterparts of $\mathbf{A}$ and $\mathbf{B}$, defined by a time parameter Δ . To overcome the input-invariant limitations of SSM, Mamba [11] makes $\mathbf{B}$, $\mathbf{C}$, and Δ input-dependent via linear projection, enhancing interactions along the sequence. In particular, the hardware-aware algorithms (e.g., parallel scan, kernel fusion, and recomputation) are employed for efficient computation [23, 41]. In this study, we harmonize the merits of Mamba with feature selection schemes, proposing the GMMamba model to address computational complexity and local redundancy interference, leading to improved performance in bag prediction.

3.2. Overview of GMMamba

Figure 3 illustrates our proposed GMMamba network. Our primary objective is to effectively capture global dependencies while mitigating redundancy interference for more accurate bag prediction.

Instead of random groupings, which fail to consider spatial and structural similarities between neighboring tissue instances and make it more difficult to establish instance relationships, we propose a location-based clustering scheme.

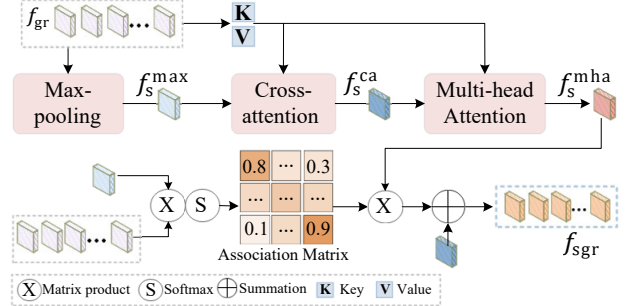


Figure 4. Illustration of our proposed cross-group super-feature sampling (CSS) module.

This method partitions each bag $\{f_{\text{ins}}^b\}_{b=1}^B$ into G groups $\{f_{\text{ins}}^g = \{f_m^g\}_{m=1}^M \mid g \in [1, G = \frac{B}{M}]\}$, where M is the number of instances in a group and g denotes the g -th group. To effectively model instance dependencies while alleviating computational complexity and redundancy, we propose an IMM module, consisting of a bidirectional Mamba (BiMamba) block and an attention block. The former captures semantic and spatial features within a group f_{ins}^g , while the latter selects the most informative instances based on their attention scores. These selected instances are then passed into the next BiMamba block to generate a refined group representation f_{gr}^g . Considering the scattered nature of tumor instances and the inadequate inter-group interactions that lead to sparse group representations, we devise a CSS module. This module extracts and aggregates discriminative tumor features across all groups $f_{\text{gr}} = \{f_{\text{gr}}^g\}_{g=1}^G$, producing discriminative and comprehensive “super-feature” group representations $f_{\text{sgr}} = \{f_{\text{sgr}}^g\}_{g=1}^G$. To further enhance the global representation of the bag, we employ multi-head attention mechanisms equipped with a class token (cls) to explore global relationships among $\{f_{\text{sgr}}^g\}_{g=1}^G$ to generate a consolidated class-level feature f_{cls} . Finally, f_{cls} is passed through an MLP to generate accurate bag predictions $\hat{Y}$.

3.3. Intra-group Masking Mamba

To effectively and efficiently reduce redundant instance computation and interference, our GMMamba design location-based grouping and IMM modules. The first step uses the K-Means clustering technique, which takes instance coordinates as input to generate groups $\{f_{\text{ins}}^g\}_{g=1}^G$. Within each group f_{ins}^g , we apply the BiMamba module to explore complex relationships among instances. To further filter out uninformative instances, an attention block is employed to mask $M \times M_r$ instances with low attention scores (AttentionMask), where M_r denotes the mask ratio. The informative instances $\{f_m^g\}_{m=1}^{M \times (1-M_r)}$ are then processed through the BiMamba and attention block to form the group representation f_{gr}^g . The procedures are expressed as:

$$\{\bar{f}_m^g\}_{m=1}^M = \text{BiMamba}(\{f_m^g\}_{m=1}^M), \quad (5)$$

$$\{\bar{f}_m^g\}_{m=1}^{M \times (1-M_r)} = \text{AttentionMask}(\{\bar{f}_m^g\}_{m=1}^M, M_r). \quad (6)$$

$$f_{gr}^g = \text{Attention} \left(\text{BiMamba} \left(\{ \bar{f}_m^g \}_{m=1}^{M \times (1-M_r)} \right) \right). \quad (7)$$

3.4. Cross-Group Super-Feature Sampling

The goal of our CSS module is to capture dispersed tumor information across groups for comprehensive and discriminative representations, as illustrated in Figure 4. To achieve this, CSS first employs Max-Pooling to extract the most salient features within $f_{gr}^g = \{f_{gr}^g\}_{g=1}^G$, forming an initial “super-feature” $f_s^{\max}$. The $f_s^{\max}$ is then used as a query for cross-attention to aggregate discriminative information from across groups, yielding f_s^{ca} . Considering that a single softmax score may not effectively aggregate complex inter-group relationships, we employ a multi-head attention (MHA) mechanism to refine the super-feature into a more robust representation f_s^{mha} . The processes can be summarized as follows:

$$f_s^{\max} = \text{Max-Pooling} \left(\{f_{gr}^g\}_{g=1}^G \right), \quad (8)$$

$$f_s^{ca} = \text{Cross-Attention} \left(f_s^{\max}, \{f_{gr}^g\}_{g=1}^G \right), \quad (9)$$

$$f_s^{mha} = \text{MHA} \left(f_s^{ca}, \{f_{gr}^g\}_{g=1}^G \right). \quad (10)$$

While f_s^{mha} can capture better global relations, f_s^{mha} may lose critical local details within individual groups. Hence, we compute the association matrix $\mathbf{Q} = \{Q^g\}_{g=1}^G$, which measures the relationship between $f_s^{\max}$ and $\{f_{gr}^g\}_{g=1}^G$. Based on $\mathbf{Q}$, we bridge local and global information, ensuring comprehensive super-feature group representations $f_{sgr}^g = \{f_{sgr}^g\}_{g=1}^G$ for complex WSIs. The steps are described as follows:

$$\{Q^g\}_{g=1}^G = \text{Softmax} \left(f_s^{\max} \times \{f_{gr}^g\}_{g=1}^G \right), \quad (11)$$

$$\{f_{sgr}^g\}_{g=1}^G = f_s^{ca} + \{Q^g\}_{g=1}^G \times f_s^{mha}. \quad (12)$$

4. Experiments

To thoroughly evaluate the performance of our GM-Mamba model, we conduct extensive experiments against mainstream MIL methods on four widely used WSI datasets, including TCGA Breast Cancer (BRCA) [36], TCGA Esophageal Cancer (ESCA) [36], Breast Adenocarcinoma Subtypes (BRACS) [2] and TCGA-Lung Cancer [36]. We further validate the effectiveness of location-based grouping and cross-group super-feature sampling (CSS) methods by integrating these components into classic MIL frameworks. Ablation studies assess the individual contributions of key components, including the grouping strategies, masking ratios in intra-group masking Mamba (IMM), and CSS variations.

4.1. Datasets

TCGA Breast Cancer Dataset. We collect 952 breast slides from the TCGA-BRCA project, comprising 749

WSIs for Invasive Ductal Carcinoma (IDC) and 203 WSIs for Invasive Lobular Carcinoma (ILC). Following [22], the dataset is randomly split into training, validation, and test sets with ratios of 65:10:25. To prepare the data for training, we use CLAM [26] to crop each WSI into non-overlapping 256×256 patches at $10 \times$ magnification, yielding approximately 3.1 million instances.

TCGA Esophageal Cancer Dataset. This dataset consists of 156 diagnostic WSIs, including 90 squamous cell carcinoma and 66 adenocarcinoma WSIs [36]. In line with MuRCL [55], we partition the data into training, validation, and test sets with a ratio of 3:1:1. Non-overlapping 256×256 patches at $20 \times$ magnification are extracted using CLAM [26], producing approximately 0.5 million instances.

Breast Carcinoma Subtyping Dataset. BRACS consists three categories: 265 benign, 89 atypical, and 193 malignant breast tumors [2]. Adhering to the official split [2, 46], we use 395 WSIs for training, 65 for validation, and 87 for testing. We use CLAM [26] to extract 256×256 instances at $10 \times$ magnification (approximately 1.4 million instances).

TCGA-Lung Cancer Dataset. This dataset [36] comprises 541 lung adenocarcinoma (LUAD) and 512 squamous cell carcinoma (LUSC) slides from 956 unique cases. Following the settings in previous works [21, 35, 44], we split the data using a 65:10:25 ratio. CLAM [26] generates non-overlapping 256×256 patches at $20 \times$ magnification (approximately 4.1 million instances).

4.2. Implementation Details

Following previous works [21, 22, 55], for the TCGA-BRCA, TCGA-Lung, and TCGA-ESCA datasets, we use a ResNet18 [12] encoder pre-trained on ImageNet [8] (Resnet18-ImageNet) to extract 512-dimensional features. For the BRACS dataset, we use publicly available features from [46], combining 512-dimensional ResNet18-ImageNet features and 384-dimensional ViT-S/16-SSL features (DINO-pretrained [4]). The model is optimized using bag-level cross-entropy loss and the AdamW optimizer [25], with a weight decay of $1e - 5$ and an initial learning rate of $1e - 4$. We train the GMMamba model for 100 epochs with a batch size of 1 (*i.e.*, one bag per batch) on a single NVIDIA RTX 4090 GPU.

4.3. Baseline and Evaluation Metrics

Baseline. We compare our proposed GMMamba against representative MIL methods, including attention-based (ABMIL [17], CLAM [26], DSMIL [1], MHIM-ABMIL [35], IBMIL-ABMIL [21], ILRA-MIL [39], ACMIL [46]), Transformer-based (TransMIL [31], DTFD [44], MHIM-TransMIL [35]), and Mamba-based (SSMMIL [10], MambaMIL [42]). All baselines are reproduced using their official implementations.

Table 1. Quantitative comparison of our results and MIL methods on TCGA-BRCA and TCGA-ESCA datasets using Resnet18-ImageNet feature extractor. The numbers in **red** and **blue** indicate the best and second performance.

Methods	TCGA-BRCA			TCGA-ESCA		
	Accuracy	AUC	F1	Accuracy	AUC	F1
ABMIL [17]	0.862±0.025	0.882±0.038	0.915±0.015	0.827±0.092	0.914±0.066	0.859±0.079
DSMIL [1]	0.823±0.021	0.820±0.033	0.892±0.014	0.808±0.065	0.882±0.084	0.833±0.062
CLAM-MB [26]	0.865±0.020	0.890±0.029	0.917±0.014	0.821±0.078	0.902±0.088	0.843±0.075
CLAM-SB [26]	0.858±0.011	0.877±0.029	0.914±0.006	0.834±0.061	0.927±0.064	0.861±0.049
TransMIL [31]	0.847±0.021	0.846±0.036	0.905±0.013	0.796±0.101	0.895±0.083	0.831±0.083
DTFD-MaxMin [44]	0.816±0.023	0.810±0.033	0.885±0.013	0.834±0.110	0.881±0.145	0.875±0.074
DTFD-AFS [44]	0.823±0.028	0.824±0.034	0.892±0.017	0.872±0.054	0.911±0.046	0.890±0.050
DTFD-MaxS [44]	0.828±0.038	0.826±0.049	0.891±0.026	0.777±0.116	0.820±0.092	0.827±0.081
MHIM-ABMIL [35]	0.858±0.004	0.883±0.020	0.912±0.001	0.859±0.082	0.940±0.046	0.889±0.058
MHIM-TransMIL [35]	0.848±0.022	0.872±0.013	0.905±0.012	0.853±0.054	0.911±0.040	0.879±0.044
ILRA-MIL [39]	0.857±0.035	0.886±0.026	0.908±0.027	0.841±0.098	0.901±0.091	0.857±0.089
IBMIL-ABMIL [21]	0.859±0.018	0.897±0.028	0.913±0.012	0.859±0.115	0.915±0.086	0.878±0.103
SSMMIL [10]	0.863±0.006	0.896±0.032	0.916±0.005	0.809±0.092	0.910±0.069	0.838±0.078
MambaMIL [42]	0.868±0.017	0.878±0.032	0.917±0.009	0.821±0.098	0.908±0.074	0.838±0.092
ACMIL [46]	0.869±0.017	0.900±0.019	0.920±0.009	0.885±0.078	0.948±0.042	0.901±0.067
GMMamba (Ours)	0.891±0.013	0.906±0.016	0.932±0.008	0.949±0.029	0.970±0.033	0.955±0.025

Evaluation Metrics. We use the widely adopted performance metrics from [1, 21, 50] for model evaluation, including the area under the receiver operating characteristic curve (AUC), accuracy, and F1 score (F1) with a threshold set at 0.5. Following established protocols [35, 42, 43], we use 5-fold cross-validation to address class imbalance for reliable evaluation. For each experiment, the training and validation data are split at the patient level according to the specified ratios for model selection. The mean classification performance and standard deviation across the five test sets are reported.

4.4. Comparison with State-of-the-Art Methods

We evaluate the performance of our GMMamba model against representative methods on four widely used WSI classification datasets. The quantitative results are presented in Tables 1 and 2. As shown, GMMamba outperforms all baseline methods in all metric. For the TCGA-BRCA and TCGA-Lung datasets, GMMamba surpasses attention-based methods, such as ACMIL [46] and MHIM-ABMIL [35], by 2.2%, 1.2%, 1.0%, and 0.8% in accuracy and F1, respectively. Particularly on the TCGA-ESCA dataset, GMMamba excels at outstanding accuracy, AUC, and F1, showing improvements of 6.4%, 2.2%, and 5.4%, respectively. To further validate the effectiveness of GMMamba, we conduct a three-class classification experiment on the BRACS dataset, employing two different encoders to assess the robustness of our GMMamba. Our model outperforms the classical CLAM [26] and ABMIL [17] methods, achieving improvements of 3.9% and 2.8% in terms of accuracy. Although ILRA-MIL [39] emphasizes cross-instance correlation modeling, it struggles to achieve

a discriminative and comprehensive representation, resulting in a 4.6% lower accuracy when using the ResNet18-ImageNet encoder. In particular, compared to the Mamba-based MambaMIL [42], GMMamba exhibits excellent performance, with average gains in accuracy and F1 of 7.15% and 8.2% on both encoders, respectively. The primary reason for these improvements is that most comparative methods suffer from inadequate exploration of instance correlations within the bag and local redundancy, leading to suboptimal performance. In addition, we observe that the group-based DTFD-AFS [44] outperforms bag-label methods in the TCGA-ESCA dataset but lags behind our method by 7.7%, 5.9%, and 6.5% in accuracy, AUC, and F1, respectively. These results further demonstrate the achievements of our IMM and CSS in reducing redundancy and enhancing inter-group interactions for better bag representation.

4.5. CSS Module Generalizability

To further validate the generalizability and scalability of our proposed CSS, we integrate CSS with five MIL frameworks and evaluate them on the TCGA-ESCA dataset. Quantitative results in Table 3 reveal that CSS significantly improves bag-level methods by 3.9%, 2.6%, 5.1%, and 1.9%, respectively, due to its ability to fully explore instance relationships. In particular, CSS-DTFD outperforms DTFD-AFS [44] by 0.7% in accuracy and 0.3% in F1, demonstrating CSS’s ability to aggregate dispersed tumor features for discriminative global representations.

4.6. Ablation Studies

Validation on Basic Components. We conduct ablation studies to assess the contributions of individual compo-

Table 2. Quantitative comparison of our results with MIL results on the BRACS and TCGA-Lung datasets.

Methods	BRACS				TCGA-Lung	
	Resnet18-ImageNet		ViT-S/16-SSL		Resnet18-ImageNet	
	Accuracy	F1	Accuracy	F1	Accuracy	F1
ABMIL [17]	0.691±0.041	0.604±0.055	0.791±0.048	0.715±0.082	0.844±0.023	0.849±0.021
DSMIL [1]	0.657±0.026	0.555±0.016	0.736±0.044	0.644±0.051	0.783±0.041	0.789±0.033
CLAM-MB [26]	0.689±0.036	0.601±0.024	0.747±0.038	0.684±0.045	0.844±0.023	0.849±0.021
CLAM-SB [26]	0.739±0.052	0.668±0.060	0.760±0.057	0.700±0.050	0.834±0.030	0.838±0.029
TransMIL [31]	0.706±0.044	0.596±0.036	0.767±0.029	0.671±0.042	0.819±0.038	0.823±0.032
DTFD-MaxMin [44]	0.698±0.030	0.610±0.044	0.760±0.046	0.687±0.057	0.832±0.031	0.833±0.034
DTFD-AFS [44]	0.676±0.056	0.614±0.054	0.776±0.038	0.707±0.049	0.852±0.020	0.855±0.021
DTFD-MaxS [44]	0.708±0.052	0.614±0.094	0.756±0.032	0.678±0.034	0.764±0.010	0.762±0.019
MHIM-ABMIL [35]	0.715±0.035	0.624±0.039	0.754±0.033	0.650±0.031	0.867±0.031	0.872±0.027
MHIM-TransMIL [35]	0.689±0.026	0.613±0.016	0.752±0.025	0.670±0.047	0.832±0.035	0.831±0.044
IBMIL-ABMIL [21]	0.702±0.040	0.607±0.045	0.773±0.040	0.688±0.057	0.816±0.027	0.821±0.025
ILRA-MIL [39]	0.732±0.076	0.650±0.094	0.773±0.050	0.702±0.070	0.823±0.035	0.828±0.041
SSMMIL [10]	0.721±0.037	0.620±0.048	0.760±0.056	0.676±0.062	0.843±0.033	0.847±0.034
MambaMIL [42]	0.706±0.066	0.636±0.071	0.748±0.042	0.646±0.064	0.856±0.027	0.864±0.022
ACMIL [46]	0.698±0.041	0.633±0.044	0.773±0.023	0.692±0.035	0.844±0.023	0.849±0.021
GMMamba (Ours)	0.778±0.025	0.699±0.037	0.819±0.022	0.747±0.049	0.877±0.020	0.880±0.018

Table 3. Quantitative evaluation of the CSS scheme with four MIL methods on the TCGA-ESCA dataset. $\uparrow$ **Bold** indicates an improvement over previous results. We set $G = 10$ and $M_r = 0$.

Methods	Accuracy	F1
ABMIL [17]	0.827±0.092	0.859±0.079
CSS+ABMIL	0.866±0.086 ($\uparrow$ 3.9%)	0.883±0.074 ($\uparrow$ 2.4%)
TransMIL [31]	0.796±0.101	0.831±0.083
CSS+TransMIL	0.822±0.135 ($\uparrow$ 2.6%)	0.840±0.127 ($\uparrow$ 0.9%)
DTFD-AFS [44]	0.872±0.054	0.890±0.050
CSS+DTFD	0.879±0.072 ($\uparrow$ 0.7%)	0.893±0.067 ($\uparrow$ 0.3%)
SSMMIL [10]	0.809±0.092	0.838±0.078
CSS+SSMMIL	0.860±0.099 ($\uparrow$ 5.1%)	0.876±0.097 ($\uparrow$ 3.8%)
MambaMIL [42]	0.821±0.098	0.838±0.092
CSS+MambaMIL	0.840±0.064 ($\uparrow$ 1.9%)	0.854±0.062 ($\uparrow$ 1.6%)

nents, including location-based grouping (LG), IMM, and CSS, to the overall performance. For simplicity, we start with a baseline model that aggregates the whole bag of instances using a two-layer BiMamba with Max-Pooling (w BMP). To verify the effectiveness of our LG, we compare two models: w BMP and w LG-BMP. Furthermore, to evaluate the contributions of IMM and CSS in eliminating local redundancy and facilitating inter-group interactions, we design three further models: w/o Masking, w IMM, and w CSS, where w/o Masking denotes the removal of the masking scheme from the IMM and CSS modules.

The quantitative results listed in Table 4 demonstrates that our complete GMMamba model significantly outperforms its incomplete variants. Specifically, the LG strategy facilitates BiMamba’s instance dependency modeling by clustering more relevant instances in a group, boosting accuracy and F1 improvements by 5.2%, 2.2% and 3.7%, respectively, when comparing w BMP and w LG-BMP. In addition, MHA with f_{cls} enhances inter-group communica-

Table 4. Validation of basic components on the TCGA-ESCA dataset. We set $G = 10$ and $M_r = 20\%$. The gray denotes the settings used in our method.

Model	LG	IMM	CSS	Accuracy	AUC	F1
w BMP	$\times$	$\times$	$\times$	0.833±0.056	0.903±0.053	0.862±0.046
w LG-BMP	$\checkmark$	$\times$	$\times$	0.885±0.057	0.925±0.057	0.899±0.048
w/o Masking	$\checkmark$	$\times$	$\times$	0.898±0.051	0.930±0.047	0.907±0.051
w IMM	$\checkmark$	$\checkmark$	$\times$	0.924±0.055	0.939±0.052	0.930±0.055
w CSS	$\checkmark$	$\times$	$\checkmark$	0.936±0.028	0.965±0.032	0.945±0.025
GMMamba	$\checkmark$	$\checkmark$	$\checkmark$	0.949±0.029	0.970±0.033	0.955±0.025

Table 5. Ablation analysis for cross-group feature sampling (CSS) on the TCGA-ESCA dataset. We set $G = 10$ and $M_r = 20\%$.

Model	CMax	CCA	CMHA	Q	Accuracy	F1
w CMax	$\checkmark$	$\times$	$\times$	$\times$	0.885±0.057	0.899±0.048
w CMax $\times$ Q	$\checkmark$	$\times$	$\times$	$\checkmark$	0.904±0.106	0.920±0.089
w CCA	$\checkmark$	$\checkmark$	$\times$	$\times$	0.911±0.046	0.917±0.050
w CCA $\times$ Q	$\checkmark$	$\checkmark$	$\times$	$\checkmark$	0.917±0.064	0.930±0.054
w CMHA	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	0.917±0.048	0.926±0.045
w CMHA $\times$ Q	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	0.930±0.062	0.940±0.053
w CSS (Ours)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	0.949±0.029	0.955±0.025

tion, leading to a 1.3% accuracy improvement of w/o Masking over w LG-BMP. Due to local redundancy within the group, w/o Masking lags behind w IMM by 2.6% and 2.3% in accuracy and F1. On the other hand, the CSS scheme effectively aggregates dispersed tumor information across groups for discriminative and comprehensive group representations, yielding improvements of 2.5% in accuracy, 3.1% in AUC, and 2.5% in F1 (comparing GMMamba and w IMM). While CSS significantly contributes to the improvement of model inference accuracy by 3.8% (comparing w CSS and w/o Masking), w CSS model falls behind our GMMamba by 1.3%, due to uninformative instances impairing the accuracy of the global representation.

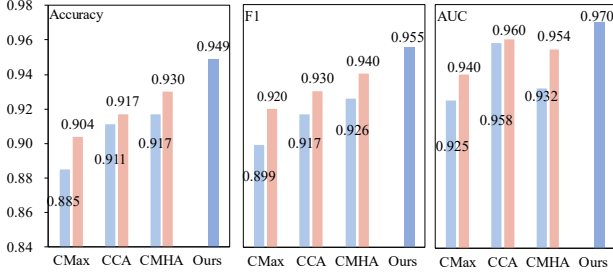


Figure 5. Ablation analysis for cross-group feature sampling (CSS) on the TCGA-ESCA dataset. The pink bar represents the outputs of the w CMax, w CCA, and w CMHA models for the super-features, each of which integrates the association matrix $\mathbf{Q}$.

Table 6. Analyzing the effects of different group methods on GMMamba. We set $G = 10$ and $M_r = 20\%$.

Methods	Accuracy	AUC	F1
Random	0.898 ± 0.072	0.934 ± 0.061	0.914 ± 0.062
Feature-based	0.911 ± 0.072	0.946 ± 0.052	0.921 ± 0.066
Local-based (Ours)	0.949 ± 0.029	0.970 ± 0.033	0.955 ± 0.025

Table 7. Effects of grouping number (G) for our GMMamba on the TCGA-ESCA dataset. M_r is set to 0.

G	Accuracy	AUC	F1
2	0.860 ± 0.091	0.890 ± 0.083	0.876 ± 0.081
5	0.917 ± 0.044	0.933 ± 0.035	0.931 ± 0.036
10	0.936 ± 0.028	0.965 ± 0.032	0.945 ± 0.025
20	0.910 ± 0.037	0.948 ± 0.032	0.926 ± 0.030

CSS Variants. To further investigate the impact of our CSS, Table 5 and Figure 5 present quantitative comparisons against different super-feature sampling schemes, including cross-group Max-Pooling (w CMax in Eq. (8)), cross-group cross-attention (w CCA in Eq. (9)), and cross-group multi-head attention (w CMHA in Eq. (10)), where the super-feature group representation f_{sgr} is denoted as $\times \mathbf{Q}$. Notably, our CSS consistently outperforms its variants, with methods incorporating the association matrix $\mathbf{Q}$ showing significant advantages over those that do not use $\mathbf{Q}$. These results demonstrate that $\mathbf{Q}$ effectively bridges local and global communication for better group representation. Furthermore, the performance of w CMax, w CCA, and w CMHA improves progressively, indicating that our CSS successfully extracts dispersed and sparse tumor features across groups, leading to better bag predictions.

Grouping Strategies. The grouping methods are crucial for identifying and reducing redundant instances. As reported in Table 6, our location-based grouping strategy demonstrates remarkable superiority, surpassing random grouping (RG) and instance feature-based clustering grouping (FCG) by 5.1% and 3.8% in accuracy. Since RG and FCG disrupt intra-group instance homogeneity, they struggle to effectively address the redundancy challenge. While FCG outperforms RG, it fails to surpass our GMMamba. Because it prioritizes high-dimensional feature similarity, it ignores spatial structure relationships among instances.

Table 8. Effects of the mask ratio (M_r) for our GMMamba on the TCGA-ESCA and TCGA-Lung datasets. G is set to 10.

M_r (%)	TCGA-ESCA		TCGA-Lung	
	Accuracy	F1	Accuracy	F1
0.0	0.844	0.872	0.792	0.798
1.0	0.875	0.895	0.802	0.798
5.0	0.875	0.900	0.830	0.842
10.0	0.938	0.941	0.797	0.814
15.0	0.875	0.895	0.816	0.822
20.0	0.938	0.947	0.807	0.806
30.0	0.906	0.914	0.807	0.818

Hyperparameter Analysis. To investigate the impact of the grouping number G and mask ratio M_r , we conduct extensive experiments on GMMamba. The results in Tables 7 and 8 illustrate that both G and M_r substantially improve classification performance. The main reason is the clustering of instances with similar tissues and structures into a group, which helps GMMamba reduce redundant and uninformative instances, leading to more precise bag representations. Furthermore, we observe that integrating the masking scheme greatly boosts model accuracy, yielding improvements of 9.4% and 3.8% in accuracy, and 7.5% and 4.4% in F1, compared to $M_r = 0$.

5. Conclusion

In this study, we propose a novel group masking Mamba (GMMamba) framework to facilitate global modeling and reduce redundancy for improved WSI classification. The core innovation of GMMamba is the intra-group masking Mamba (IMM), which addresses local redundancy and enhances group representations. To further improve bag predictions, we introduce a cross-group super-feature sampling (CSS) module to extract and aggregate dispersed tumor information, leading to more comprehensive and discriminative bag representations. Extensive experiments on multiple benchmark datasets demonstrate that GMMamba outperforms state-of-the-art methods, achieving significant accuracy improvements. While GMMamba effectively eliminates local redundancy and promotes global representations, its performance is limited by applying the same masking ratio across different groups. Future work could explore learnable ratio networks to optimize this setting. In addition, more effective instance selection methods warrant further investigation.

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